ASTR 302 Spring 2026
Introduction to Observational Astronomy

Professor Brittany E. Miles
Steward Observatory

Agenda – August 24, 2026

- Introductions
- Class Goals and Big Picture Motivation
- Syllabus
- Observing
- Questions

Introductions

- Name, Year, Major, and Pronouns
- What interests you about observational astronomy?
- What do you like to do outside of astronomy?

Instructor: Dr. Brittany E. Miles (she / her)

Education

B.S. Physics and Minor in Geophysics and Planetary Physics, UCLA
M.S. Astronomy and Astrophysics, UC Santa Cruz
PhD Astronomy and Astrophysics, UC Santa Cruz

Research

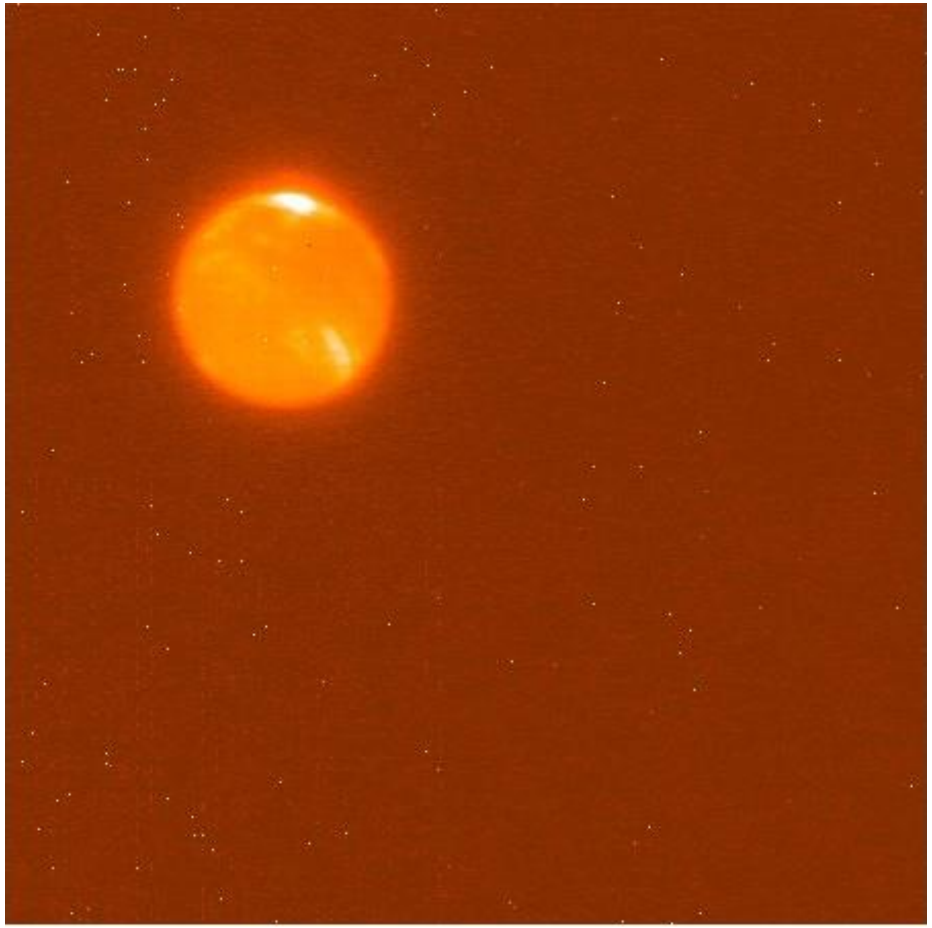
I study brown dwarfs and exoplanets with infrared spectroscopy. I also build instruments.

I was one of the first people to work with scientific data from JWST.

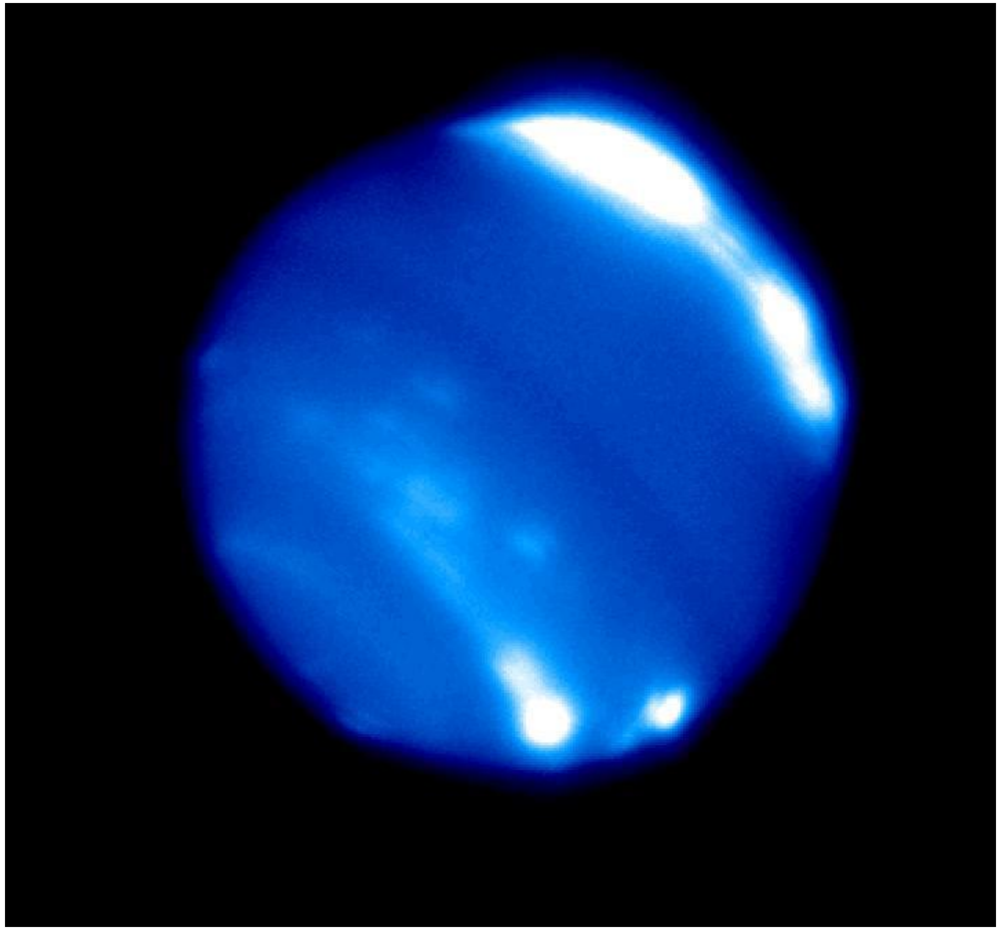
I have personally observed using Keck Observatory and the Large Binocular Telescope.

Has observational programs using Gemini Observatory, IRTF, LBT, JWST

Class Goals: Manipulate Data from a Telescope to Get Scientifically Useful Information



Raw Image of Neptune Keck/NIRC2



Reduced Image of Neptune Keck/NIRC2 from the "Twilight Zone" Program

Class Goals: Plan, Understand, and Communicate Why Observations are Important

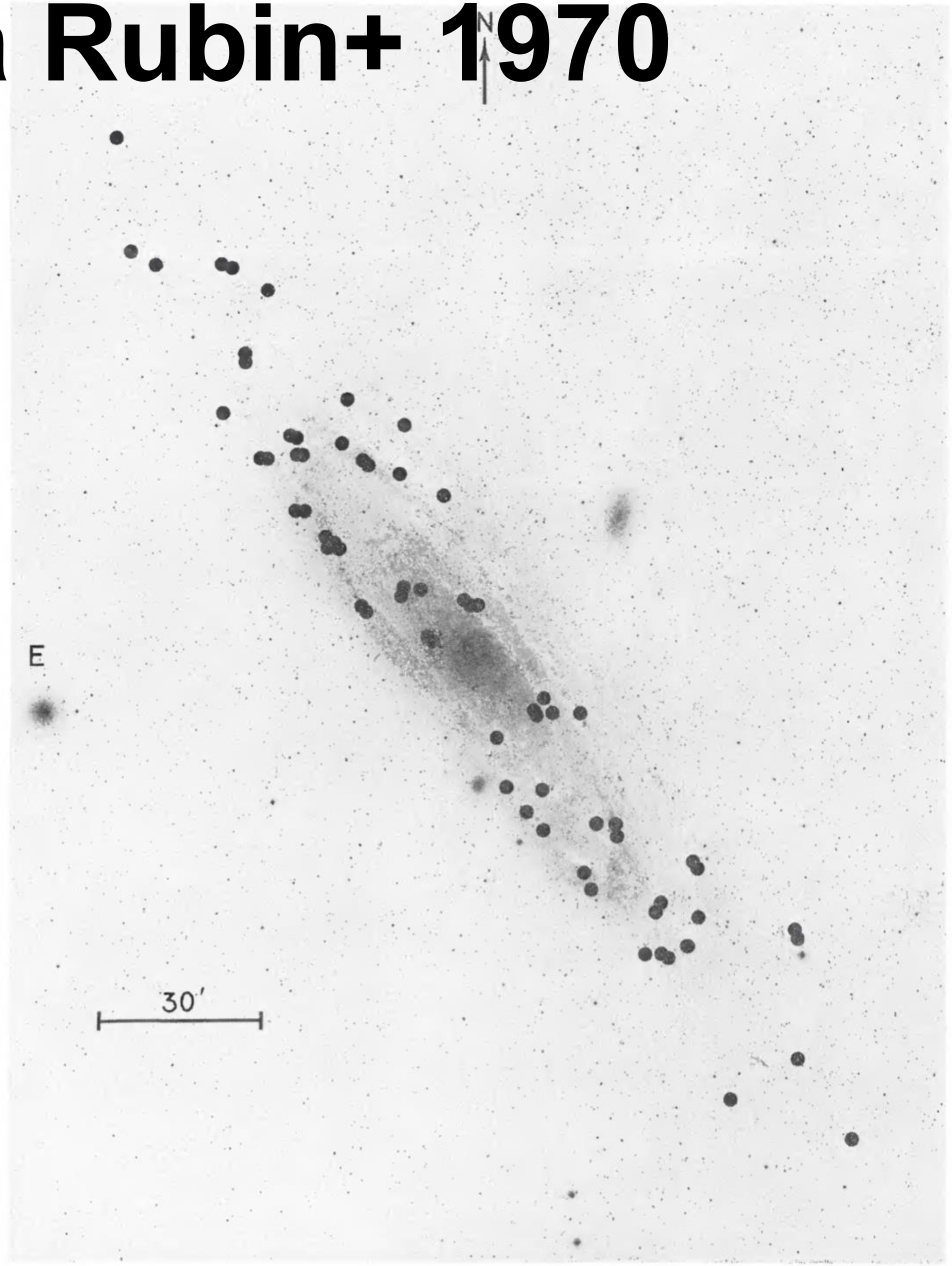
- Learn how to pose a scientific question and answer it with a specific telescope and instrument.

- Comprehend why **imaging** or **spectroscopic** data looks the way it does.

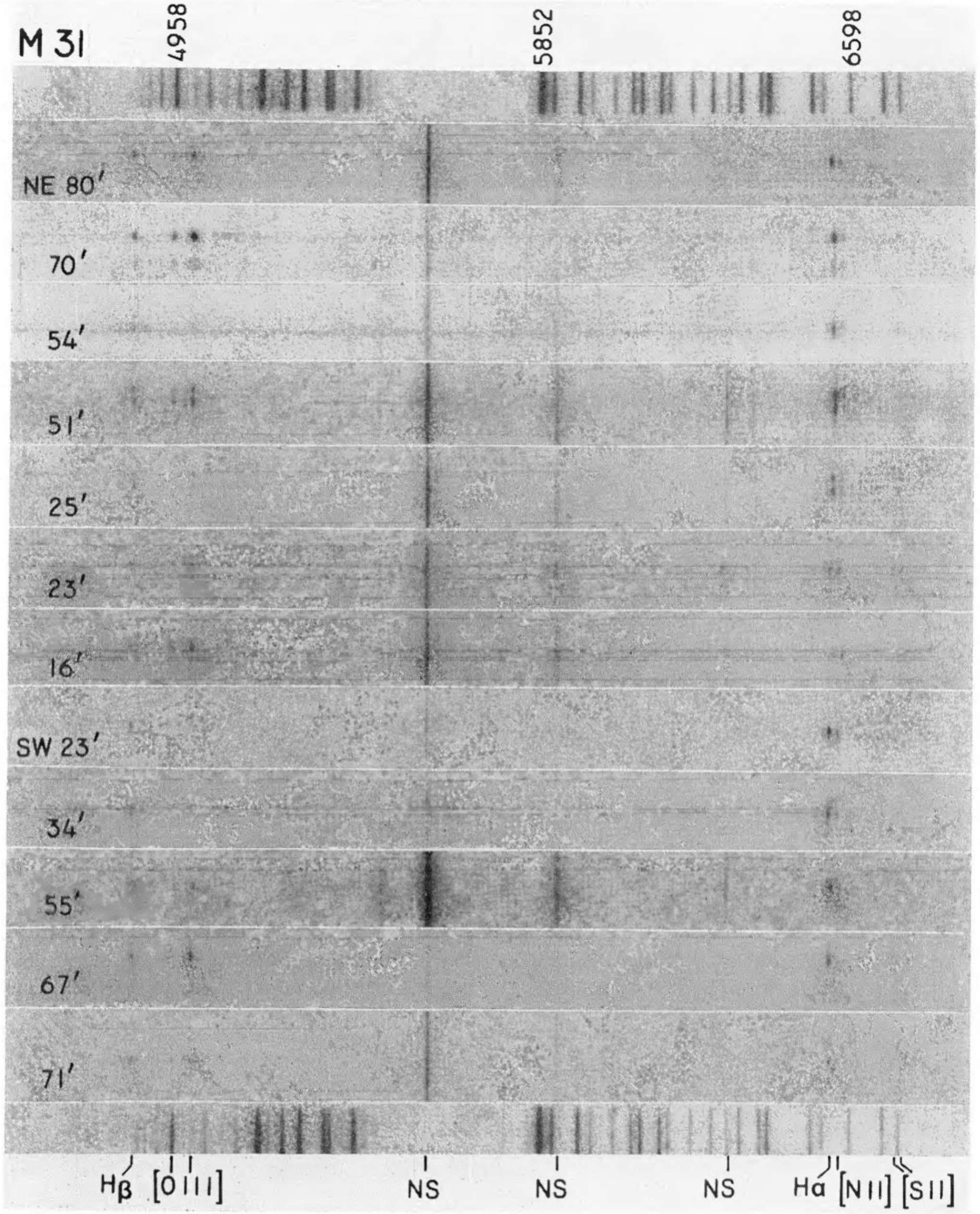
- Communicate to others your methods, results, statistical significance, and their importance.

- Observations are not only used to confirm existing theories, they push the boundaries of how we understand the universe.

Vera Rubin+ 1970

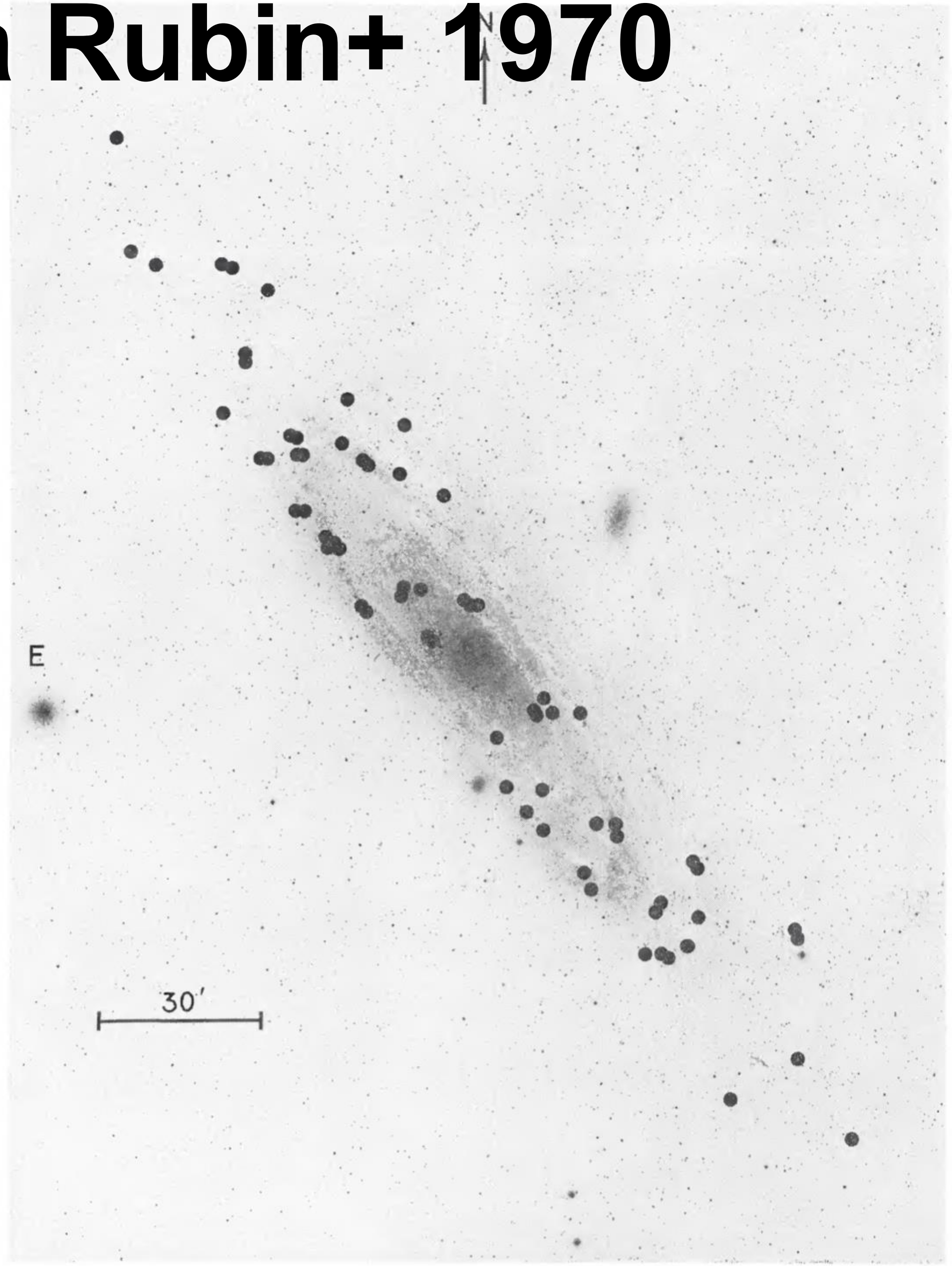


M31 Imaging



Spectroscopy

Vera Rubin+ 1970



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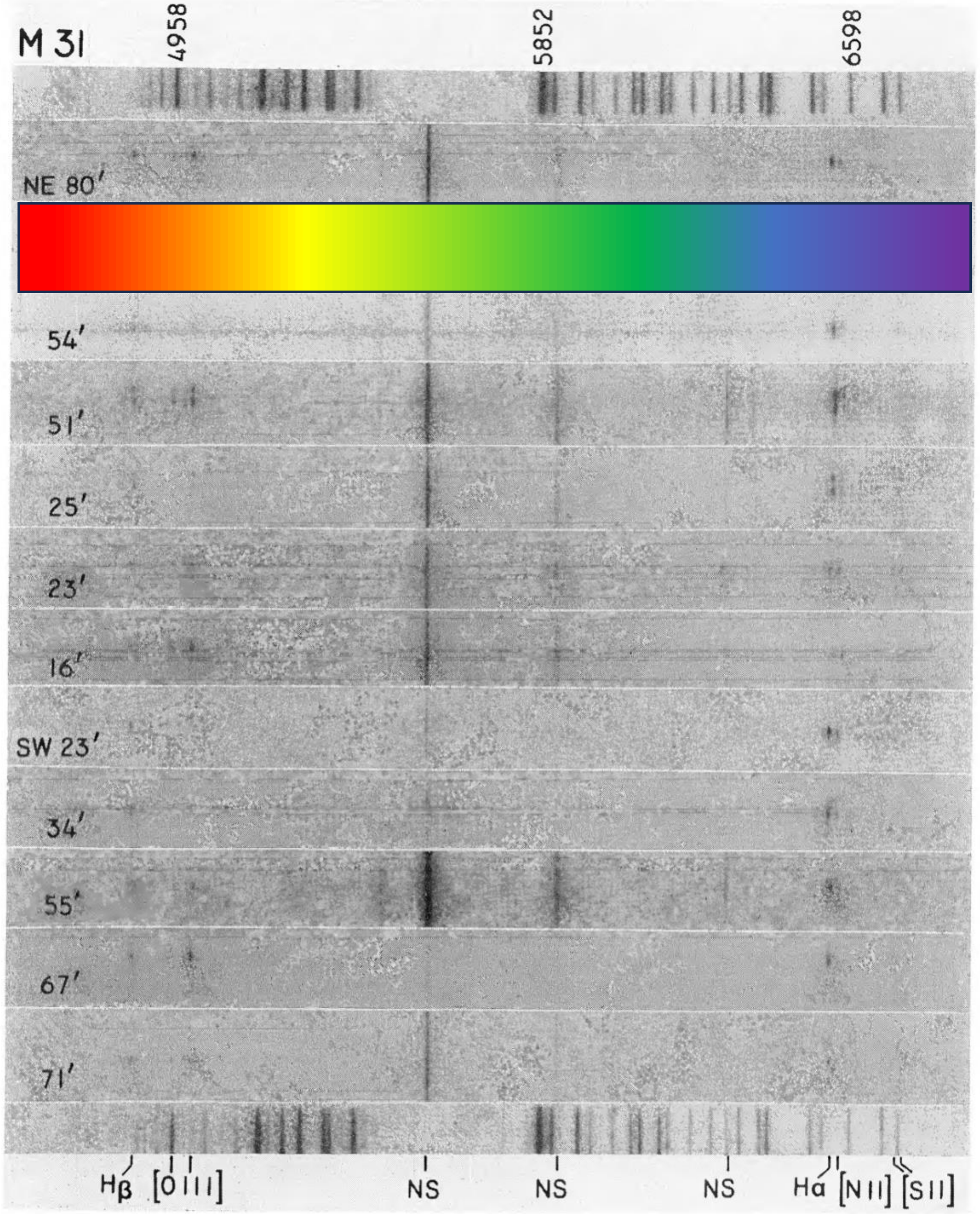
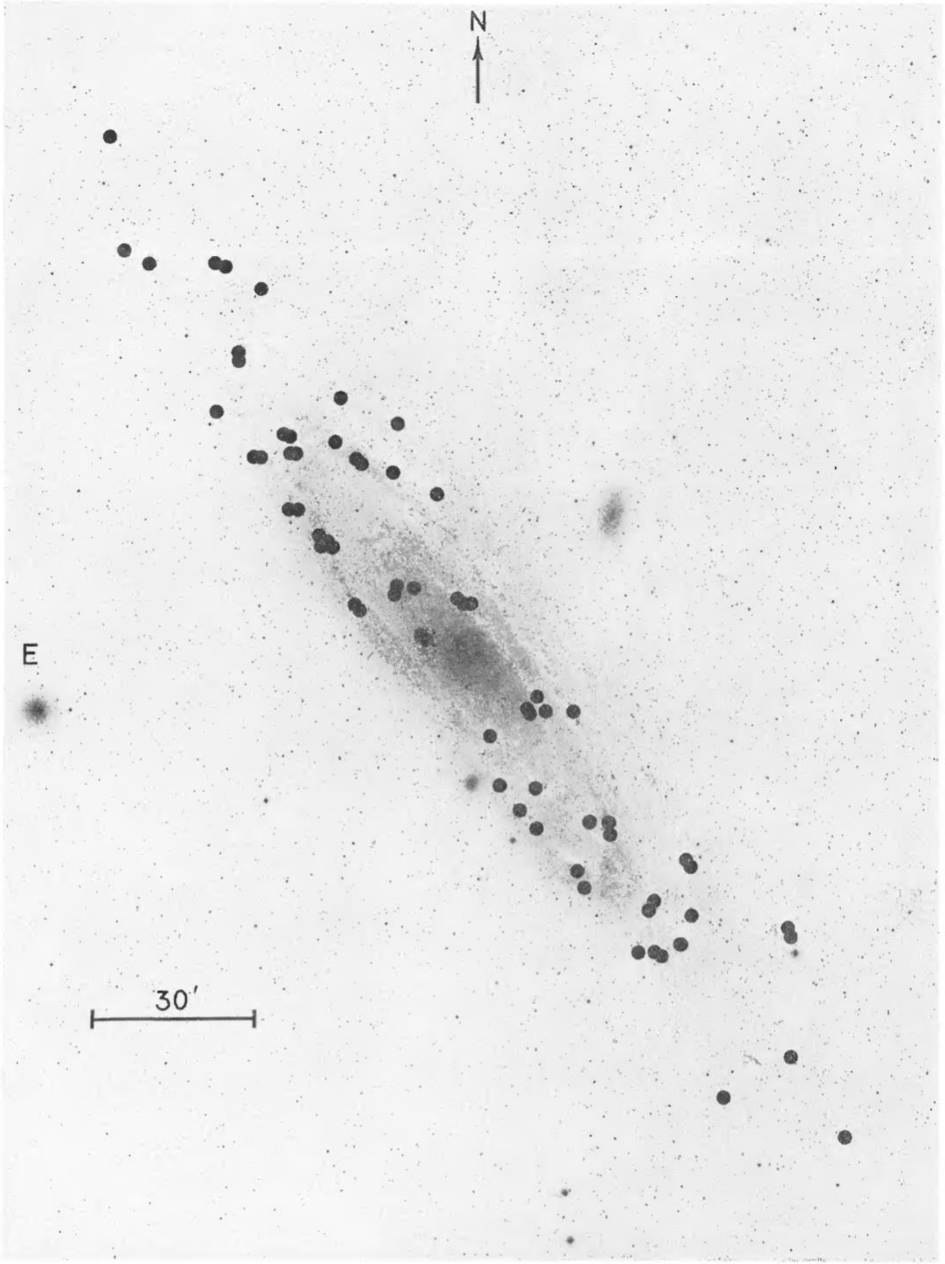


Spectroscopy

Vera Rubin+ 1970



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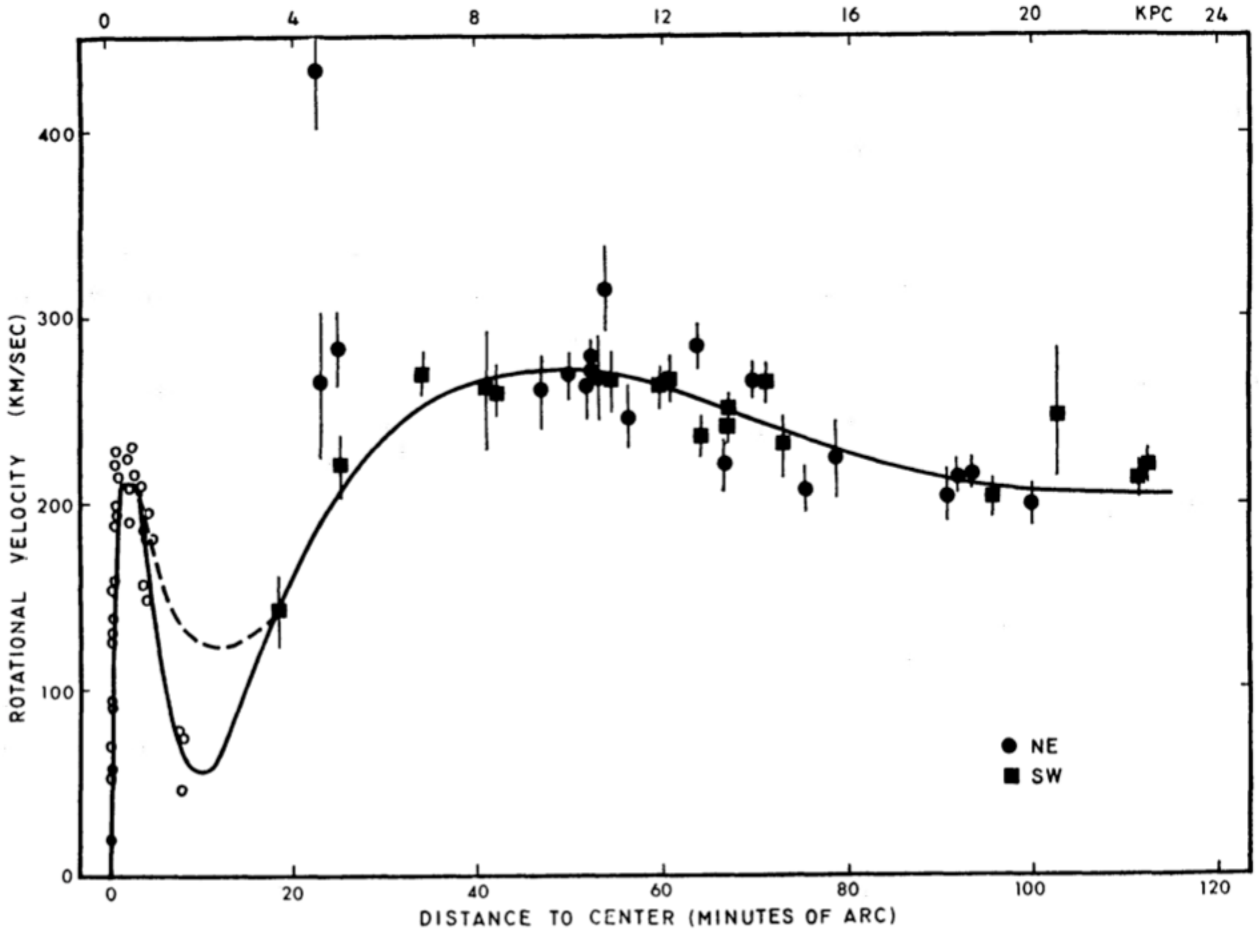
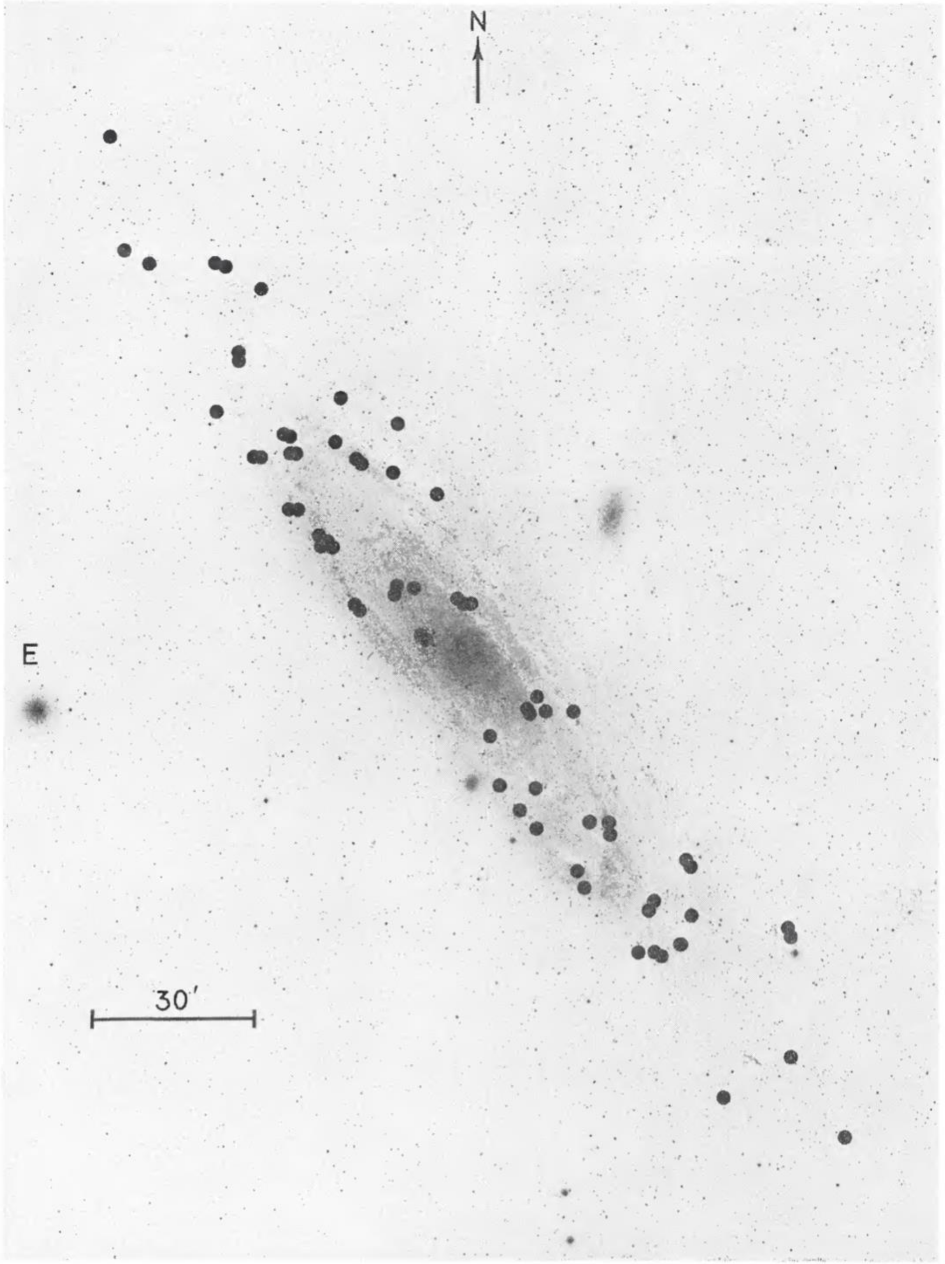
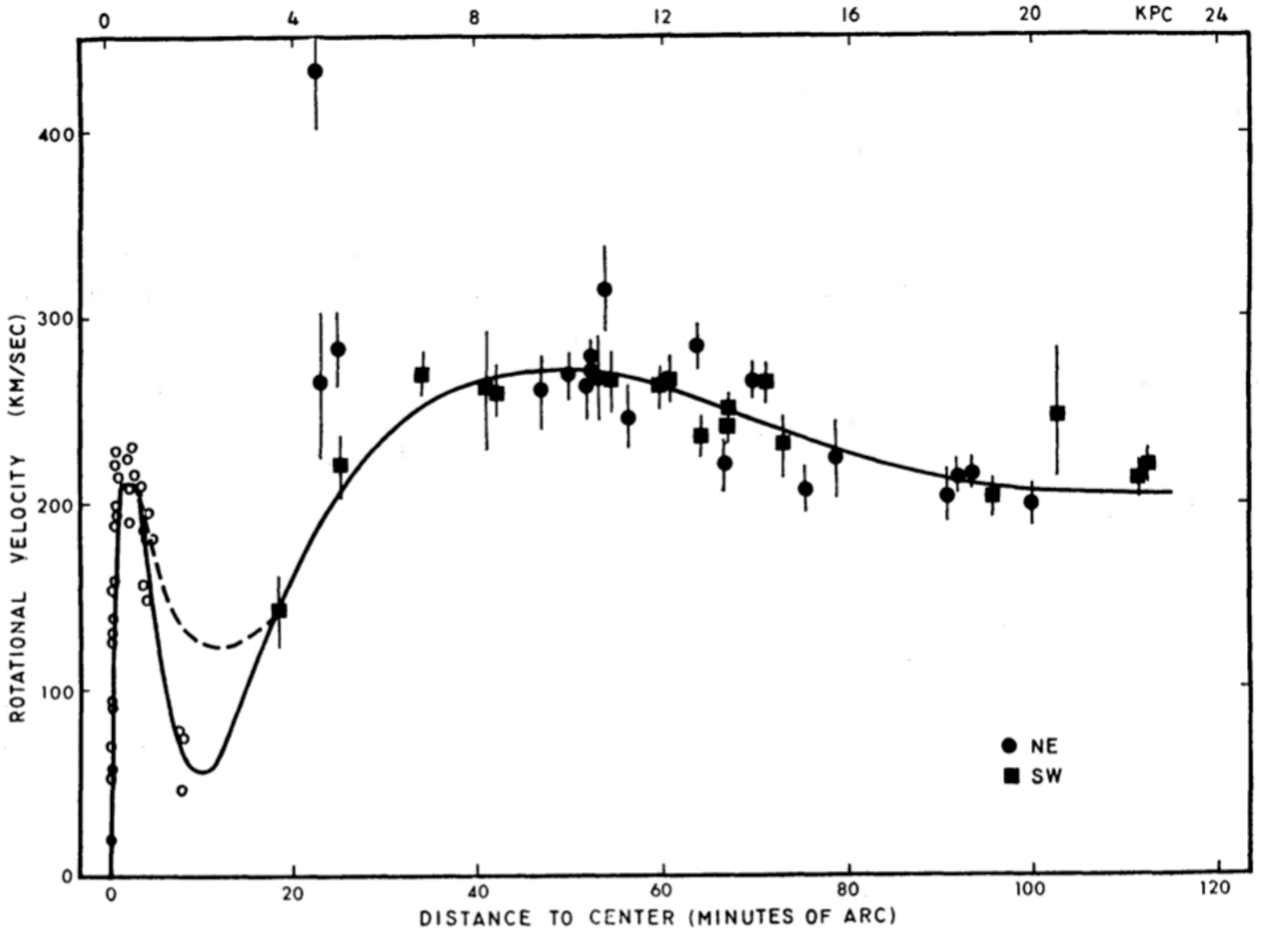


Velocity curve of emission regions measured with spectroscopy.

Vera Rubin+ 1970



M31 Imaging



The velocity of the emission regions is too high given the amount of visible matter -> dark matter

Syllabus Discussion: Communication

- Discord (research groups, Dr. Miles, Evan, Jialin, Zach)
- Email (Dr. Miles, Evan, Jialin, Zach, Dr. Milne)
- D2L – Course website is here (Class Demo)

Syllabus Discussion: Class Structure

- This course is generally structured so that Mondays and Wednesday are more lecture and problem solving based.
- Friday you work with python, your research group, or go to the telescope.
- Homework is meant to reinforce planning, python-based coding, or written skills necessary for your final project. **The final project is a marathon not a race.**

Syllabus Discussion: Final Project

- The **Final Project** is supposed to be executed over the entire semester. Do not leave things until the last minute.

- I will assign you into groups of ~4 based on interests.

- You will need to consult Dr. Milne and I on how to execute your project and if its feasible.

  - Friday, We'll talk about what and how to observe. Read Telescope Procedures by Friday at 10am.
  - Week 2, I will assign groups
  - Week ~3, your group will present your project idea

Syllabus Discussion: Final Project Logistics

- On select days, a group goes up to the telescope to take data for their project.
  - Available weekends: 10/2 (two groups), 10/9, 10/16 (two groups), 10/31
  - Available weekdays: 9/29 – 10/1 (Tuesday)
- Your TAs will drive your group to Mt. Bigelow or Kitt Peak for the observing trip.
- Keep up regular communication with your research groupmates. Maintain a collegial environment and think about how to do the project equitably. At the end of your final reports you will document who did what for the project.
- You are expected to write an individual report at the end of the semester. Your group will present together at the end of the semester.

Logistical Challenges with A302

- This is one of the largest A302 classes to date. (the major is growing)
- I cannot control the weather + climate change happening.
- Our TAs are also TAs for another class. (COS funding)
- I have decided to cancel some classes to maximize the likelihood of you getting the chance to observe with telescopes.